



False memory confidence depends on the prefrontal reinstatement of true memory

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ABSTRACT

For confidence of memory, a neural basis such as traces of stored memories should be required. However, because false memories have never been stored, the neural basis for false memory confidence remains unclear. Here we monitored the brain activity in participants while they viewed learned or novel objects, subsequently decided whether each presented object was learned and assessed their confidence levels. We found that when novel objects are presented, false memory confidence significantly depends on the shared representations with learned objects in the prefrontal cortex. However, such a tendency was not found in posterior regions including the visual cortex, which may be involved in the processing of perceptual gist. Furthermore, the confidence-dependent shared representations were not observed when participants correctly answered novel objects as non-learned objects. These results demonstrate that false memory confidence is critically based on the reinstatement of high-level semantic gist of stored memories in the prefrontal cortex.

1. Introduction

Memory confidence crucially affects how we make decisions in our daily lives. Memory confidence generally depends on the accuracy of memory, but dissociation between memory confidence and accuracy has also been reported (Moritz et al., 2006b; Simons et al., 2010). A typical case of such dissociation is false memory with high confidence (Jou and Flores, 2013; Kim and Cabeza, 2007a). In these cases, people are convinced of the accuracy of their memory, despite the fact that they have never experienced it. Confidence in false memories can unintentionally lead to incorrect decisions or can cause trouble for an innocent person, such as in the case of false testimony in court (Wade et al., 2010; Wells and Olson, 2003).

How is confidence generated for false memories that have never been stored in the brain? Traditionally, false memories are considered to depend on semantic or perceptual relatedness (Bergman and Roediger, 1999; Deese, 1959; Koutstaal et al., 2003, 1999; Koutstaal and Schacter, 1997; Roediger and McDermott, 1995; Wickens, 1970) or on source-misattribution (Mitchell and Johnson, 2009; Ost et al., 2013; Van Damme and Smets, 2014; Zhu et al., 2013). In particular, the paradigm most commonly used in false memory research is the Deese–Roediger–McDermott (DRM) paradigm (Roediger and McDermott, 1995; Ruci et al., 2009; Wang et al., 2019; Ye et al., 2016; Zhu et al., 2019), and numerous studies using the DRM or similar modified

paradigms suggest that false memories occur due to shared semantic or perceptual gist between a retrieval probe and stored memory traces (Jou and Flores, 2013; Kim and Cabeza, 2007a; Moritz et al., 2006a; Ye et al., 2016; Zhu et al., 2019). Thus, according to this view, the confidence level of a false memory may be based on how much information shared with related true memory is evoked during false recognition.

Neuroimaging studies have shown common neural activities between target and lures across the prefrontal cortex (PFC) using DRM or similar modified paradigms (Kim and Cabeza, 2007b; Okado and Stark, 2003; Pidgeon and Morcom, 2016). These findings propose that the overlapping neural activities may reflect shared neural processing across true and false memories in the PFC (Kim and Cabeza, 2007b; Okado and Stark, 2003). In particular, given that the PFC plays a key role in semantic processing, including generalization and semantic integration (Badre et al., 2005; Goldberg et al., 2007; Hayama and Rugg, 2009; Pan and Sakagami, 2012), numerous studies suggest that the common involvement of the prefrontal cortical areas may reflect common properties or shared semantic meaning such as the category between the target and related lures. The overlap in the neural activity of the dorsolateral prefrontal cortex (dlPFC) has been interpreted as the retrieval of semantic labels (Blacker et al., 2006; Hayama and Rugg, 2009; Hoenig and Scheef, 2009; Miotto et al., 2006). Common activities in the dlPFC and anterior prefrontal cortex (aPFC) are also thought to be associated with the retrieval of shared contextual information (Müsch et al.,

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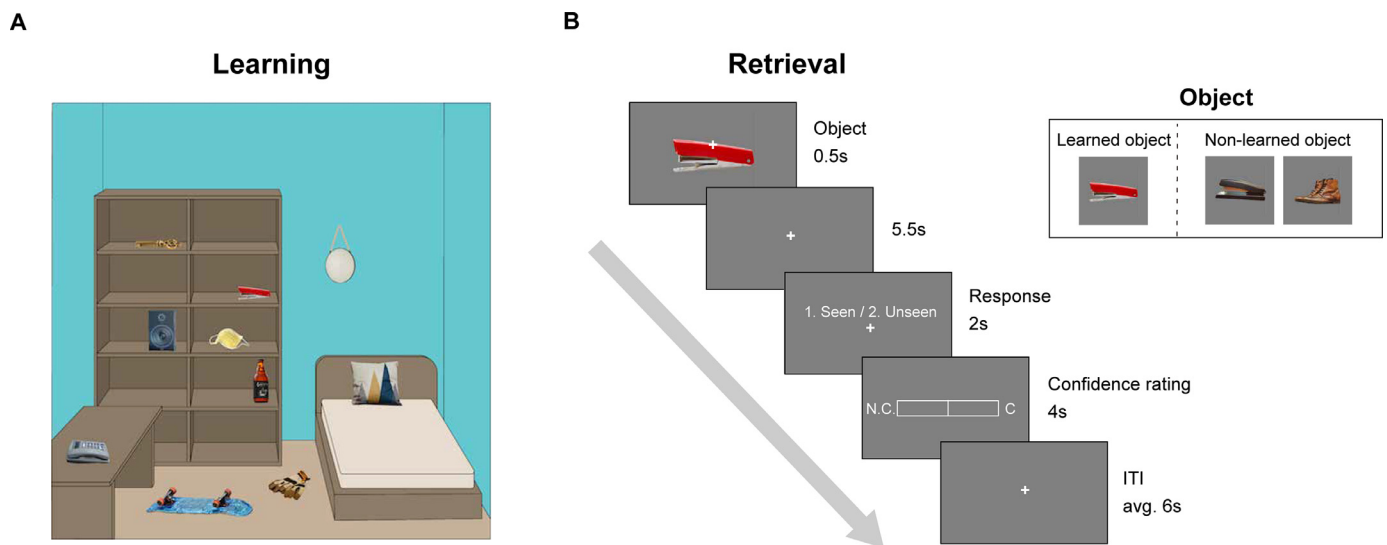


Fig. 1. Experimental design. A. An example of a room image. During each trial of the learning runs, a participant viewed a room image containing ten objects. B. During each trial of the retrieval runs, an object image was presented on the screen, followed by delay, response, and confidence rating phases. During the response phase, participants had to respond as to whether the object presented before the delay was included in any of the learned room images between the “Seen” and “Unseen” choices. There were three types of object stimuli: learned objects, non-learned objects that belong to the same basic-level category as any learned object, and non-learned objects that did not belong to any basic-level category of learned objects. During the confidence rating phase, the participants were instructed to report their confidence rating for the response by moving the center bar left or right. The leftmost end indicated “not confident” (N.C., level 0) and the rightmost end indicated “confident” (C, level 100), and there were eleven confidence levels from 0 to 100 in intervals of 10. During the retrieval runs, Korean words were actually used, but the English version is shown here for readers. The Korean version that was presented to participants is shown in supplementary Fig. 1.

2020; Rajah et al., 2011, 2010; Ramnani and Owen, 2004; Simons et al., 2005). Moreover, response changes in the aPFC and dlPFC have been linked to memory confidence levels in various types of recognition tests (Fleck et al., 2006; Maniscalco and Lau, 2012; Murray and Ranganath, 2007; Shekhar and Rahnev, 2018). Therefore, it may be plausible that confidence for false memories is based on the prefrontal reinstatement of the high-level components, such as semantic gist, of stored true memory. However, common activities in PFC regions have often been interpreted as a reflection of general retrieval processing, such as control or monitoring (Kouneiher et al., 2009; McGuire and Botvinick, 2010; Miller and Cohen, 2001; Sohn et al., 2007; Walton et al., 2004). Moreover, overlapping neural activities between true and false memories have also been found in posterior regions, including the sensory cortex (Dennis et al., 2012; Kim and Cabeza, 2007b; Slotnick and Schacter, 2004). Therefore, these findings lead to another possibility that the common activities observed in the PFC regions may be related to the control or monitoring of the reinstatement of stored memories in the posterior regions, and the confidence level of a false memory may be critically based on the perceptual gist reinstated in the posterior regions. Thus, without any direct comparisons between information representations during true and false memory processes, the neural bases for the confidence of false memory remain unclear.

To reveal the critical neural bases for the confidence in false memory, we designed an experiment in which participants recognized learned objects and rated their confidence level for each object while their brain activity was monitored by means of functional magnetic resonance imaging (fMRI). During the experiment, learning and the subsequent retrieval runs were repeated four times. During each learning run, the participants were instructed to observe four images of a room in which everyday objects were placed (Fig. 1A). This encoding paradigm was designed to induce a wide range of memory strength levels and confidence levels for objects naturally. During the following retrieval run, they were asked to indicate whether each presented test object was seen previously with a subjective rating of confidence (scale 0–100, interval of 10) (Fig. 1B and Supplementary Fig. 1). Half of the test objects presented in the retrieval runs were learned objects and the other half

were non-learned (novel) objects. Half of the non-learned objects belonged to the same basic-level category as the learned objects and the other half were objects in different categories. Objects belonging to the same basic-level category share both the semantic gist and the perceptual gist (Jordan et al., 2015; Malone et al., 2016). Given the possibility that false memory confidence depends on the reinstatement of the high-level semantic gist of stored memory (Hebscher and Gilboa, 2016; Koutstaal and Schacter, 1997; Moritz et al., 2006b, 2006a), we focused on neural responses in the anterior prefrontal cortex (aPFC) and the dorsolateral prefrontal cortex (dlPFC). We directly compared neural representations during instance of false recognition of non-learned objects (false alarm trials) with those during the recognition of learned objects (hit trials) and tested whether the degree of shared representations is critically linked to the confidence level of false memories. Furthermore, we examined shared representations in the visual cortices considering the possibility that false memory confidence may also depend on the reinstatement of perceptual gist in the sensory cortex. Additionally, given that the hippocampus is a core region for true memory retrieval (Bookheimer et al., 2000; Brown and Aggleton, 2001) and considering that a few prior studies have indicated the involvement of the hippocampus during false memory processing (Abe et al., 2008; McTighe et al., 2010; Paz-Alonso et al., 2013; Wing et al., 2020), we also examined whether hippocampal representations are associated with false memory confidence.

2. Methods

2.1. Participants

Forty Korean participants (age 22.90 ± 3.47 years, 18 females) took part in the experiment. All participants were right-handed with normal (or corrected to normal) vision and had no history of neurological or psychiatric disorders. All participants provided written informed consent for the procedure in accordance with protocols approved by the Korea Advanced Institution of Science and Technology (KAIST) Institutional Review Board.

2.2. Stimuli

The fMRI experiment consisted of 'learning' runs and 'retrieval' runs. During the learning runs, participants were instructed to observe four room images. Each room had the same structure and the same furniture arrangement (a desk, a bed, and a bookshelf), and contained ten everyday objects (e.g., stapler, phone) (Fig. 1A). Different sets of objects were placed on the furniture for each room. Thus, 40 objects of different basic-level categories were used during the learning runs. The color of the walls was red for two rooms and was blue for the other two rooms. All of these room images were generated with Adobe Illustrator CC. The positions of the objects were manually determined such that they were evenly distributed from the central area to the periphery and naturally blended into the room image. The sizes of the objects were also manually modified to vary and were ensured to be appropriate for each type of object. The size of each room image was set to a 7° visual angle square. Differently colored rooms were designed for an additional examination of any effect on the confidence levels depending on the source memory strength. However, because no source memory (what color room each object came from) was observed (the mean accuracy of source memory was not significantly greater than the chance level), we did not analyze the source-memory-dependent effect further.

Among the ten objects present in each room image, which was presented during the learning runs, five objects were also presented during the retrieval runs. Thus, 20 objects (five objects X four rooms) from the learning runs were used as target objects during the retrieval runs. All participants learned the same room images with the same target objects. The same target objects were fixed across the learning and retrieval runs. Additionally, 20 new object images were generated and used during the retrieval runs. Among the 20 new objects, ten objects belonged to the same basic-level categories of the target objects (lures), and the other ten objects did not belong to any basic-level category of the objects used during the learning runs (foils). Two different sets of lure objects (set A and set B without overlap between the sets) were generated, with set A applied to half of the participants and set B applied to the other half. During the retrieval runs, each object image of a size characterized by a 7° visual angle was separately displayed on a uniform gray background.

All object images were selected from three websites (<http://unsplash.com>, <http://pixabay.com>, <http://www.pexels.com>) that provide freely usable, non-copyrighted images. All visual stimuli were viewed via a back-projection display (1024 × 768 resolution, 60Hz refresh rate).

2.3. Experimental design

During the fMRI experiment, the 'learning' and the subsequent 'retrieval' runs were repeated four times with the same set of objects inside an fMRI scanner. Each learning run consisted of 4 trials (one trial per room) (Fig. 1A). During each trial of the learning runs, one room image with ten objects was presented on the screen for 4 s. Within a run, each room image was presented once, and the inter-trial interval (ITI) was 10 s. During the learning runs, the participants were instructed to observe four room images and memorize the objects presented in each room image. The order of the room images within a learning run was pseudorandomized.

Each retrieval run was followed by each learning run. Each retrieval run consisted of 40 trials (one trial per object). During each trial of the retrieval run, an object image with a 7° visual angle was presented on the screen for 0.5 s. After a 5.5 s delay, two choices ("1: Seen / 2: Unseen") were given in Korean on the screen for 2 s (Fig. 1B and Supplementary Fig. 1). Participants had to respond as to whether the object was included in any of the room images learned during the learning runs within 2 s by pressing a button. After the response phase, the participants were instructed to report their confidence rating on the previous response. During this confidence rating phase, a rectangle with white edges and a white bar in the center of the rectangle is presented on the

screen for 4 s. Korean text was used to indicate the meaning of each direction ("Not confident," "Confident") at each end of the rectangle. Participants could move the bar and select among eleven horizontal positions, and they were informed that each position corresponded to a confidence level from 0 to 100 in intervals of 10. The length of each trial was 12 s, and the inter-trial interval (ITI) varied with an average ITI of 6 s. A central fixation cross was presented throughout the experiment at the center of the screen, and the participants were instructed to maintain their gaze on the central fixation cross. The order of the object stimuli was pseudorandomized.

Before the fMRI scan, participants were given instructions about the learning and retrieval runs, and went through learning and retrieval tutorials consisting of two trials of each task with stimuli not used in the main experiment.

2.4. Behavior data analysis

Due to the broad range of confidence responses, the distribution of the confidence responses varied among the participants. Therefore, to reflect the individual standards in these cases, confidence ranges for high confidence and low confidence were defined for each participant and for each type of trial. For false alarm (FA) trials from a single participant, the three confidence levels from the highest confidence level rated by the participant were defined as a high confidence range (e.g., If the highest confidence level rated by the participant is 100, the range of the high-confidence levels is 80, 90, and 100), and the three levels from the lowest confidence level rated by the participant were defined as a low confidence range. Although the criteria for the 'buffer score' between the high confidence and low confidence ranges were not specified, the mean difference between the high confidence and low confidence ranges were 36 ± 1.843 across the participants. High and low confidence ranges were also determined in the same way for correct rejection (CR) trials. When the high confidence range and the low confidence range overlapped, we excluded that participant from the fMRI data analyses. In addition, if there was not either a high- or a low-confidence level rating for the ten lure objects that belong to the same basic-level categories of the target object, that participant was also excluded from the fMRI analyses. Overall, five participants were excluded when comparing the high and low confidence trials of FA, and two participants were excluded for the analyses of CR trials. In addition to the main analyses based on individually defined confidence ranges, we also conducted the same analyses based on the absolute confidence threshold for high and low confidence levels, given the limitation of reflecting the moderate range rather than reflecting actual high or low confidence range when the confidence ranges were individually defined. Based on this absolute criterion, the high confidence range corresponds to 80, 90, and 100, and the low confidence range corresponds to 10, 20, and 30. All participants who had at least one trial for each condition (high confidence for the within-category comparison, low confidence for the within-category comparison, high confidence for the between-category comparison, and low confidence for the between-category comparison) were included, but 16 participants were excluded from the fMRI analyses because there was no high- or low-confidence level rating according to the absolute criterion.

Trials in which participants did not respond were excluded from the calculation of the confidence levels and the analyses based on the confidence levels. Trials in which participants reported a zero score for their confidence were also excluded from the analyses because we considered that such trials indicate no-confidence or 'don't know' trials. It is likely that the participants responded randomly in these trials, and these responses in FA trials are, in theory, the equivalent of a correct rejection without confidence. Nonetheless, including the trials with confidence scores of zero in our main analyses also showed the same tendency as the results without the trials. In the calculation of memory accuracy, all trials, including those without responses, were counted.

2.5. fMRI data acquisition

MRI data were acquired on a 3T Siemens MAGNETOM Prisma scanner located in the Center for Neuroscience Imaging Research of the Institute for Basic Science. Whole brain echo-planar imaging (EPI) data were acquired using a 64-channel head coil with an in-plane resolution of 2.5×2.5 mm and 48 2.5-mm slices (0.25-mm inter-slice gap, repetition time (TR) = 2 s, echo time (TE) = 17 ms, matrix size = 76×76 , field of view (FOV) = 192 mm, and flip angle = 90°). Standard MPRAGE (magnetization-prepared rapid-acquisition gradient echo) images were collected with an in-plane resolution of 1×1 mm and with 1-mm slices (TR = 2 s, TE = 2.45 ms, FOV = 256 mm, flip angle = 8°) after the experimental runs in each participant.

2.6. Regions-of-interest (ROIs)

To define the hippocampus and the ROIs of the prefrontal cortex, we used the parcellation of *Freesurfer*: “middle frontal gyrus” for the dorsolateral prefrontal cortex (dlPFC), “transverse frontopolar gyri” combined with “transverse frontopolar sulci” for the anterior prefrontal cortex (apFC), and “hippocampus” for the hippocampus. Object-selective lateral occipital complex (LOC) and retinotopic early visual cortex were determined by independent functional localizer scans. The LOC was localized with the contrast of object images versus retinotopically matched scrambled images (Lee et al., 2012), and central EVC was localized with the contrast of central (5°) and peripheral ($6\text{--}15^\circ$) flickering checkerboards (Lee et al., 2012).

2.7. fMRI data analysis

Data analysis was conducted using AFNI (<http://afni.nimh.nih.gov>, RRID:SCR_005927) version 19.0.09, SUMA (AFNI surface mapper), *FreeSurfer* (RRID:SCR_001847) version 6.0.0, and custom MATLAB scripts. EPI data were corrected for slice timing differences and head motion, after which the percent signal change was calculated for each run of fMRI scan and each participant on a voxel-by-voxel basis.

We used a standard general linear model based on the AFNI software package to deconvolve the event-related BOLD responses. Based on the AFNI program, ‘3dDeconvolve’ with ‘GAM’ option, the trials of the retrieval runs were modeled as separate events, and the β -value and t -value of each voxel were derived for the peak amplitudes of BOLD responses induced by the onset of each object presentation. For the average magnitude of responses across all voxels and objects within each ROI, β -values were used. For calculating pattern similarities or corresponding indices, we used t -values as they tend to be slightly more stable (Misaki et al., 2010). To derive corresponding indices for FA trials, we computed the Pearson correlation coefficient (r) values by comparing t -values from the voxels within each ROI for each pair of FA and Hit trials from different runs. In within-category comparison, the FA trials for the non-learned objects that belong to the same basic-level categories as the learned objects and the Hit trials for the learned objects in the matched category were compared. In between-category comparison, the FA trials for the non-learned objects that belong to the same basic-level categories as the learned objects and the Hit trials for learned objects in a non-matched category were compared. In the searchlight analysis (Kriegeskorte et al., 2006), differences in corresponding indices were derived based on the response patterns within individual spheres (radius 8.75 mm, corresponding to ~ 123 voxels).

2.8. Statistical analysis

To examine statistical comparisons between two different conditions, we used two-tailed paired t -tests without any assumptions. Specifically, we used a t -test to compare the mean confidence levels between correct and incorrect trials (Fig. 2B), for a comparison between the mean pattern similarities for the same objects and those for different objects

(Fig. 3A), to compare the mean corresponding indices for high confidence to those for low confidence (Figs. 3C, 4B, 5C, 6A, 6C, S3), to compare the mean corresponding indices for the within-category comparisons to those for the between-category comparisons (Figs. 3C, 4B, 5C, 6A, 6C, S3), and to compare the mean magnitudes of the BOLD responses for high-confidence trials with those for the low-confidence trials (Fig. S4). To compare activation, pattern similarity, or corresponding indices with basal level (zero or chance level), one-sample two-tailed t -test was used without any assumptions. Repeated-measures ANOVAs (tests of within-subjects effects) were used to determine the statistical significance of any confidence-level, category-condition, or hippocampal subregion effect. For two-way ANOVAs, Greenhouse-Geisser correlations were used. For correlation analyses for behavioral results, two-tailed Spearman’s correlation was used. To examine the contribution of different trial numbers to the corresponding indices, permutation tests were conducted. In these tests, we randomly shuffled the labels for the confidence level (high confidence and low confidence) while maintaining the number of high-confidence and low-confidence trials for each participant, and derived the outcomes from the neural responses induced by the randomized labels. We repeated this process 10,000 times and tested whether the actual differences in corresponding indices fell within the top 5% of the simulated null distribution of the differences. We used MATLAB (RRID:SCR_001622) R2021a and SPSS (RRID:SCR_002865) v26 for statistical analyses.

3. Results

3.1. Behavioral performance and confidence levels

The participants showed progressively higher memory accuracy and confidence as they learned the objects, as expected. The mean accuracy across the runs of the retrieval session was 0.656 (Run 1 = 0.559; Run 2 = 0.618; Run 3 = 0.698; Run 4 = 0.744), with a hit rate of 30.6%, a miss rate of 17.3%, a CR rate of 32.2%, and a FA rate of 15.7% (Fig. 2A). The participants reported average confidence ratings of 75.15, 52.14, and 59.63 for the hit trials, the miss trials, and CR trials, respectively (Fig. 2B). Notably, they also reported a substantial level of confidence at 54.85 for FA trials, representing the confidence level for false memories (Fig. 2B).

The average confidence was significantly higher in correct trials than in incorrect trials for both learned and non-learned objects ($t(39) = 14.547$, $p < 0.001$ for learned objects; $t(39) = 2.782$, $p = 0.008$ for non-learned objects). However, the confidence difference between correct and incorrect trials was significantly smaller for non-learned objects than for learned objects ($t(39) = 6.678$, $p < 0.001$) (Fig. 2B), and a two-way ANOVA with object type (learned, non-learned) and correctness as factors also yielded significant interaction between these factors ($F(1,39) = 44.596$, $p < 0.001$). These results indicate that memory confidence is less dependent on memory accuracy for non-learned objects than for learned objects. We also examined the relationship between individual mean accuracy and confidence levels. While there was positive correlation between memory accuracy and confidence levels for learned objects ($r = 0.615$, $p < 0.001$), no significant relationship between them was found for non-learned objects ($r = -0.113$, $p = 0.486$) (Fig. 2C). This result showed that mean confidence level of individual participants depends on their memory performance for learned objects but not for non-learned objects. Taken together, these results suggest that confidence for non-learned objects is not based simply on the process of restoring stored memory as it is.

3.2. False memory confidence depends on the shared representations with true memory in the prefrontal cortex

To test whether false memory confidence depends on the shared reinstatement of high-level components of true memory, we focused

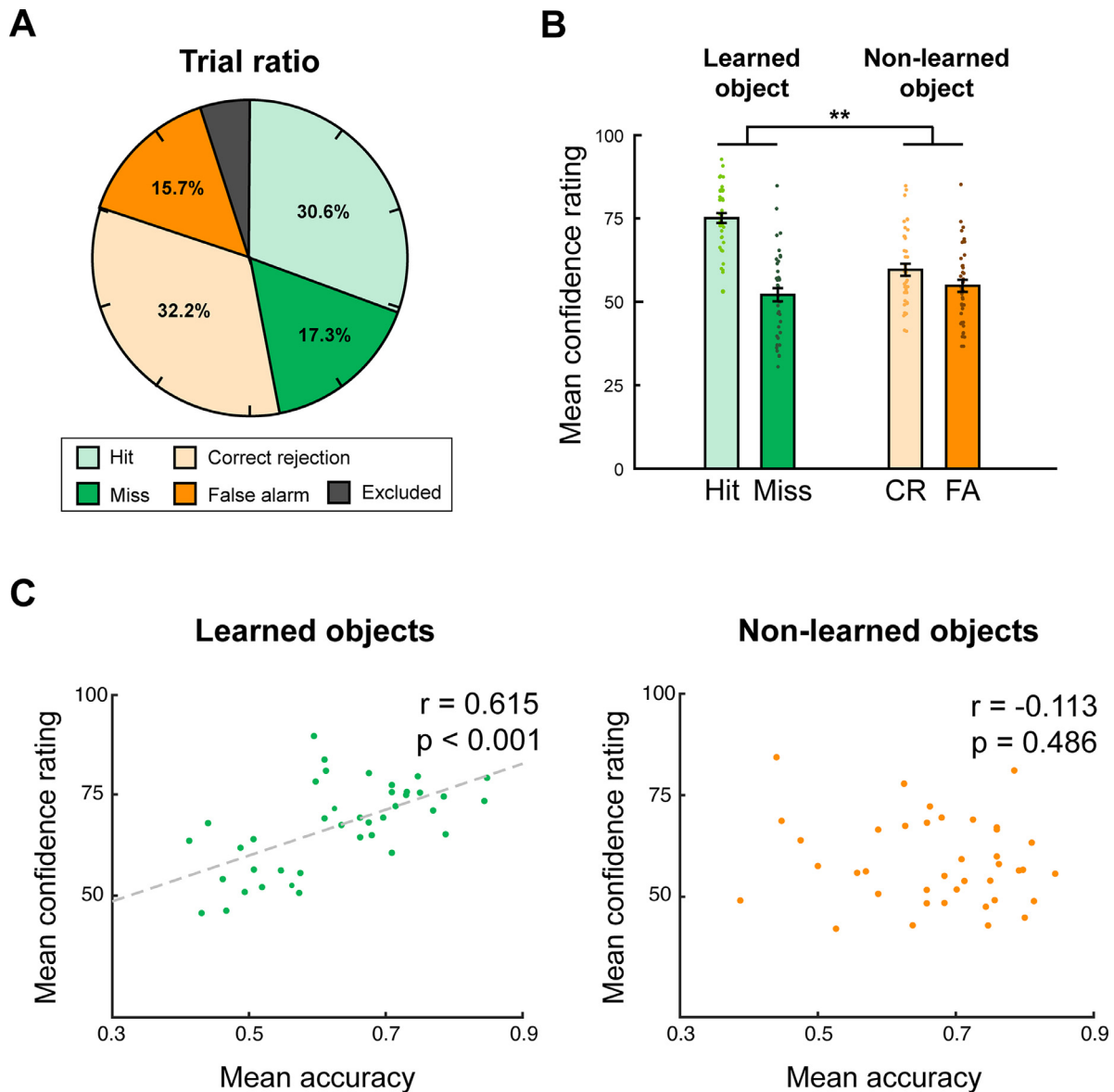


Fig. 2. Behavioral results. **A.** The ratio of trials for each type of response. There were four types of responses: correct responses for learned objects (hit), incorrect responses for learned objects (miss), correct responses for non-learned objects (correct rejection, CR), and incorrect responses for non-learned objects (false alarm, FA). Trials for which no response was given and trials with a confidence level of 0 were excluded from the fMRI data analyses. **B.** Mean confidence levels for each type of responses. Each dot indicates the mean value of the confidence ratings in each participant. Error bars indicate between-subjects s.e.m. Significant interaction between object type (learned, non-learned) and correctness was found (two-way ANOVA. ** $p < 0.01$). **C.** Relationship between the individual levels of the mean confidence rating and mean accuracy for the learned objects (left panel) and the non-learned objects (right panel).

on the prefrontal representations. First, we examined whether the prefrontal regions contain object-specific representations during true memory recognition as a prerequisite for investigating the relationship between true and false memory representations. To do this, we compared the neural pattern similarity for the same objects (within-object correlations) and that for different objects (between-object correlations) for hit trials. We found that within-object correlations were significantly stronger than between-object correlations in the aPFC and dlPFC, indicating significant discrimination of individual object representations in these regions ($t(35) = 2.332$, $p = 0.026$ for aPFC; $t(35) = 2.811$, $p = 0.008$ for dlPFC) (Fig. 3A). Thus, during recognition of truly memorized objects, individual object information was specifically processed in the prefrontal cortex.

We subsequently examined how similar false memory representations were to true memory representations in the prefrontal cortex dur-

ing object recognition. We directly compared the response patterns evoked by object recognition during FA trials with those induced by object recognition during hit trials (Fig. 3B). The prefrontal cortical regions are involved in the semantic processing of object perception (Goldberg et al., 2007; Hayama and Rugg, 2009). Thus, if the response patterns induced by the recognition of non-learned objects during FA trials correspond significantly with those induced by the recognition of learned objects during hit trials, this may indicate that during false recognition, non-learned objects are processed as learned objects based on the representation of shared semantic content in the prefrontal cortex.

The corresponding index for each FA trial was derived as a correlation value between the response patterns induced by the non-learned object of the FA trial and those evoked by the learned object of each hit trial (Fig. 3B). In particular, because objects in the same basic-level are

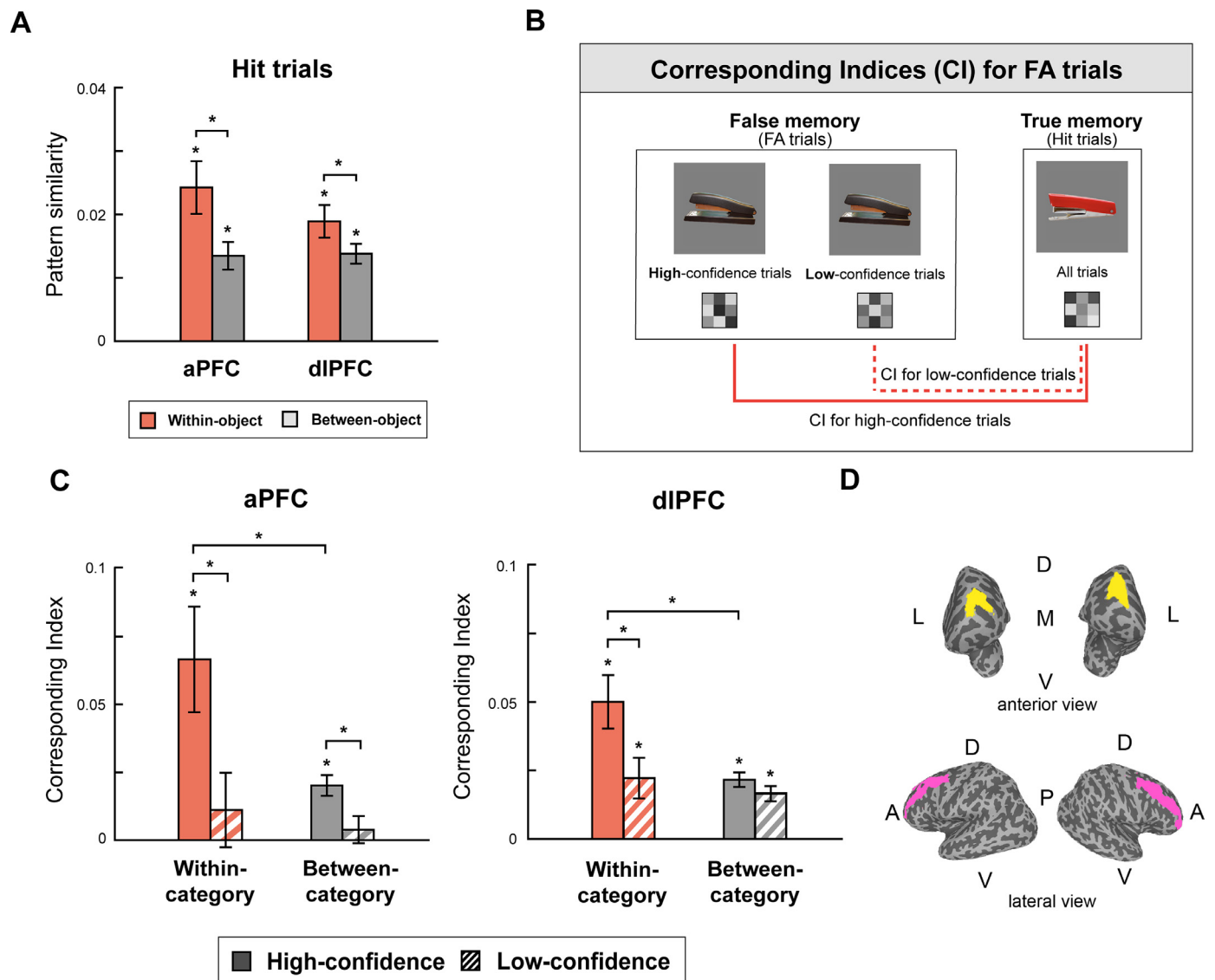


Fig. 3. Confidence-dependent shared representations with true memory during false recognition in the prefrontal cortex. **A.** Object-specific representations during true memory recognition (hit trials) in the aPFC and dlPFC. The neural pattern similarity for the same objects (within-object correlations) was significantly greater than it was for different objects (between-object correlations), indicating significant discrimination of individual object representations in the aPFC and dlPFC. **B.** Corresponding indices for FA trials were derived by direct comparisons between the response patterns evoked by object recognition during FA trials with those induced by object recognition during hit trials. The corresponding indices for high-confidence FA trials were compared with those for low-confidence FA trials. **C.** Average corresponding indices for FA trials in the aPFC and dlPFC under the within-category comparison or between-category comparison condition. The corresponding indices during the FA trials with high confidence were significantly greater than the corresponding indices during the FA trials with low confidence for the within-category comparison in both the aPFC and dlPFC. The corresponding indices during FA trials with high confidence for the within-category comparison were also significantly greater than those for the between-category comparison in both the aPFC and dlPFC. * $p < 0.05$. Error bars indicate between-subjects s.e.m. **D.** Location of aPFC (yellow) and dlPFC (pink) ROIs on inflated frontal and lateral surfaces.

semantically close, first we focused on the corresponding indices based on the comparison between the response patterns induced by learned objects during hit trials and those induced by non-learned objects in the matched basic-level categories during FA trials (within-category comparison). We found that the corresponding indices during FA trials with high confidence were significantly greater than the corresponding indices during FA trials with low confidence for the within-category comparison in the aPFC and dlPFC ($t(34) = 2.117$, $p = 0.042$ for aPFC; $t(34) = 2.347$, $p = 0.025$ for dlPFC) (Fig. 3C). The high and low confidence levels were defined relatively based on the individually normalized range of the confidence rating (see Methods). These results suggest that false memory confidence depends on the degree of shared representations between targets and lures in the prefrontal cortex.

We also examined the corresponding indices based on the comparison between the response patterns evoked by learned objects during hit trials and those for non-learned objects in non-matched categories during FA trials (between-category comparison). Even for this between-category comparison, we found significantly greater corresponding indices for FA trials with high confidence than for FA trials with low confidence in the aPFC ($t(34) = 2.410$, $p = 0.022$), but the absolute level was much lower compared to within-category comparisons ($t(34) = 2.380$, $p = 0.023$ for high confidence trials) (Fig. 3C). In the dlPFC, there was no significant difference in the corresponding indices between high-confidence and low-confidence FA trials ($t(34) = 1.653$, $p = 0.108$), and the corresponding indices during high-confidence FA trials were significantly greater for within-category comparisons compared to between-

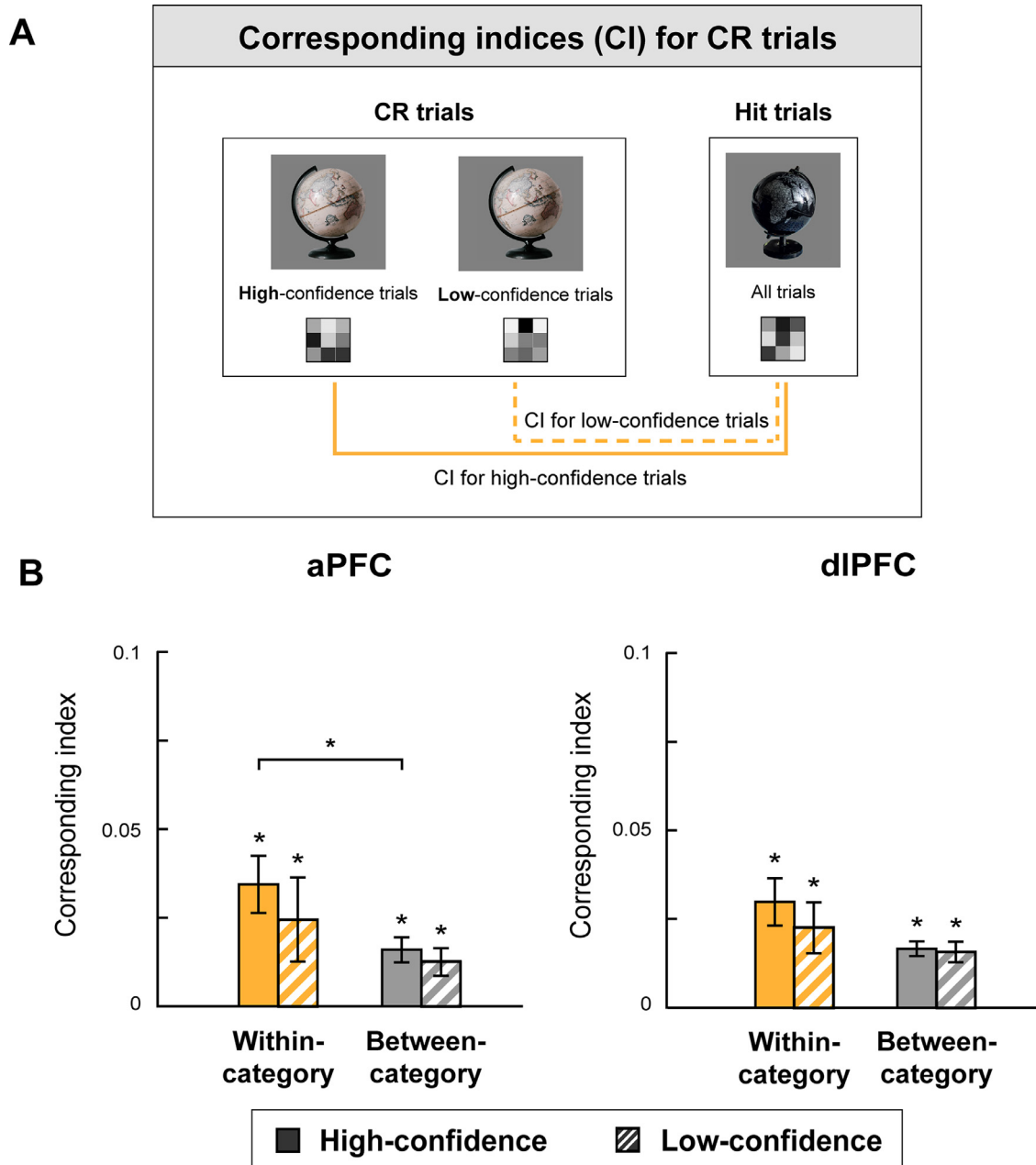


Fig. 4. No reflection of confidence in the shared representations between correct rejection and hit trials in the prefrontal cortex. **A.** Corresponding indices for CR trials were derived by direct comparisons between the response patterns evoked by object recognition during CR trials with those induced by object recognition during hit trials. The corresponding indices for high-confidence CR trials were compared with those for low-confidence CR trials. **B.** Average corresponding indices for CR trials in the aPFC and dIPFC. No significant differences were found between the corresponding indices during the CR trials with high confidence and those with low confidence in the aPFC and dIPFC for both the within-category and between-category comparisons. * $p < 0.05$. Error bars indicate between-subjects s.e.m.

category comparisons ($t(34) = 2.931$, $p = 0.006$) (Fig. 3C). Moreover, a two-way ANOVA with confidence level (high, low) and category condition (within-category, between-category) as factors yielded a highly significant main effect of confidence level ($F(1,34) = 6.425$, $p = 0.016$ for aPFC; $F(1,34) = 7.525$, $p = 0.010$ for dIPFC) as well as a main effect of category condition ($F(1,34) = 6.025$, $p = 0.019$ for aPFC; $F(1,34) = 8.539$, $p = 0.006$ for dIPFC) in both the aPFC and dIPFC (Fig. 3C). Collectively, these results show that false memory confidence-dependent shared representation is much more prominent in within-category comparison than between-category comparison, and support that the confidence level of false memory is based on shared reinstatement of semantic component of true memory in the prefrontal cortex.

Given that the number of FA trials with high-confidence rating and that with low-confidence rating were variable across the participants (Table S1), we additionally examined the possibility of whether these different trial numbers might contribute to the corresponding indices. We conducted a permutation test by randomizing the trial labels (high vs low confidence) and computing the correspondence indices from them while using the same number of high-confidence and low-confidence trials for each participant. Repetition of this approach confirmed that the stronger corresponding indices in the false alarm trials with high confidence compared to those with low-confidence are not influenced by the trial number difference (Supplementary Fig. 2A). In addition, the same analysis of the contrast between within- and between-category condi-

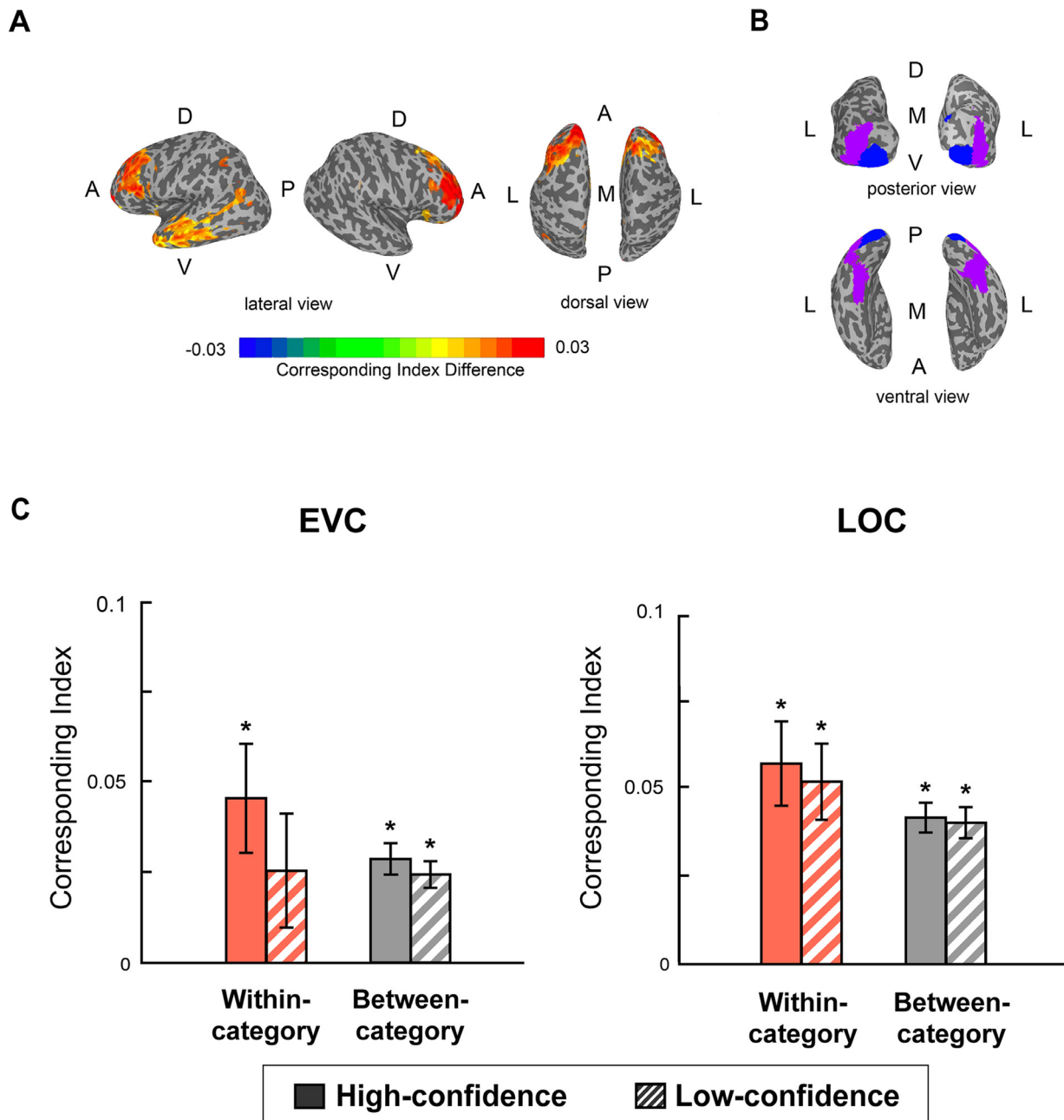


Fig. 5. No reflection of confidence in the shared representations with true memory during false recognition in the visual cortex. **A.** Searchlight results for significant differences in corresponding indices between FA trials with high confidence and FA trials with low confidence for the within-category comparison. **B.** Functional ROIs (EVC, LOC) on inflated ventral and occipital surfaces. Blue (EVC) and purple (LOC) areas indicate voxels within the ROI (EVC or LOC) in at least 15 participants. **C.** Average corresponding indices for FA trials in EVC and LOC. No significant differences were found between the corresponding indices during the FA trials with high confidence and those with low confidence in both the EVC and LOC. * $p < 0.05$. Error bars indicate between subjects s.e.m.

tions also showed no effect of the trial number difference (Supplementary Fig. 2B).

Because the high- and low-confidence levels were defined relatively based on the individually normalized confidence rating range, it might be possible that individual differences affect the corresponding indices. To clarify whether different confidence levels rather than differences in intrinsic characteristics are the main factor behind the greater corresponding indices in the FA trials with high confidence, we conducted the same analysis based on high- and low-confidence levels defined in terms of absolute rating values. Thus, the FA trials rated as 80, 90, and 100 were classified as high-confidence level trials, and the FA trials rated as 10, 20, and 30 were classified as low-confidence level trials.

Even in this comparison, the same tendency found in the comparison based on the relatively defined confidence levels was obtained (Supplementary Fig. 3A). These results confirmed that the corresponding indices in the prefrontal cortex are mainly influenced by confidence levels rather than individual differences in the range of the confidence rating.

We also compared the average magnitudes of the neural responses between high-confidence FA trials and low-confidence FA trials. We found no significant difference in the activation level between FA trials with high-confidence and those with low-confidence across the prefrontal ROIs ($t(35) = 0.568$, $p = 0.574$ for aPFC; $t(35) = 1.798$, $p = 0.081$ for dlPFC) (Supplementary Fig. 4).

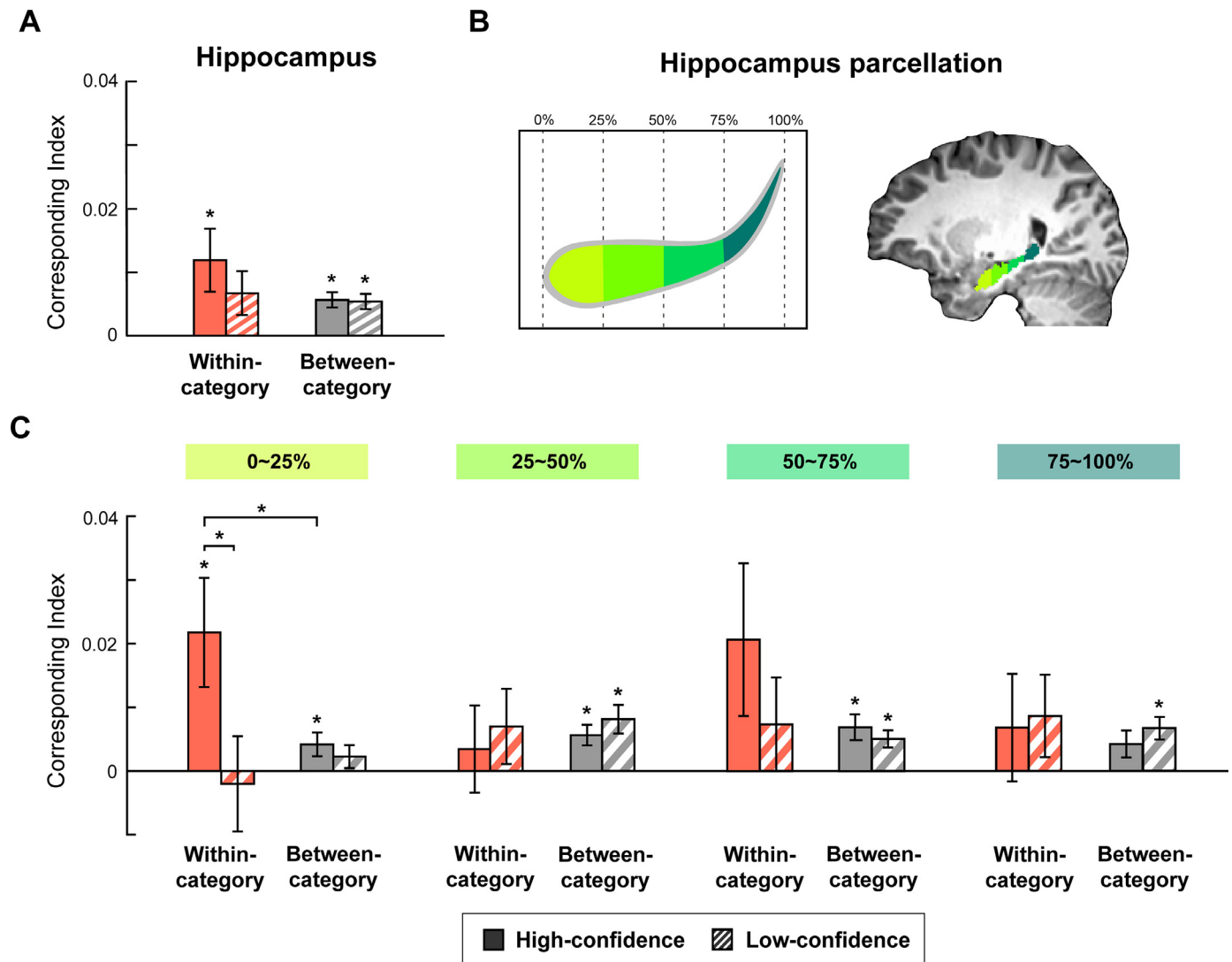


Fig. 6. Confidence-dependent shared representations with true memory during false recognition in the anterior hippocampus. **A.** Average corresponding indices for FA trials in the whole hippocampus. **B.** The hippocampus was divided into four parts (0~25%, 25~50%, 50~75%, and 75~100% in length along the anterior-posterior axis) for the subsequent analyses. **C.** Average corresponding indices for FA trials in each part of the hippocampus along the anterior-posterior axis. Only in the most anterior part (0 ~ 25%), the corresponding indices during the FA trials with high confidence were significantly greater than those with low confidence for the within-category comparison. * $p < 0.05$. Error bars indicate between-subjects s.e.m.

Taken together, our results suggest that false memory confidence is based on the degree of representation of high-level components shared with true memory in the prefrontal cortex. The prefrontal representation of more shared information with true memory may induce higher confidence in false memories.

3.3. The confidence level during correct rejection trials does not depend on shared prefrontal representations

In the CR trials, despite the fact that the non-learned objects were presented as in the FA trials, participants correctly rejected non-learned objects. This performance in CR trials may be based on discriminable representations between learned and non-learned objects, and a higher degree of shared representations in high-confidence trials may be not observed. To confirm this prediction, we derived corresponding indices based on comparisons between the response patterns evoked by non-learned objects during CR trials and those for learned objects during hit trials (Fig. 4A). As expected, we found no significant difference in the corresponding indices between the high-confidence trials and the low-confidence trials in the aPFC and dlPFC for both within-category

($t(37) = 0.624$, $p = 0.536$ for the aPFC; $t(37) = 0.758$, $p = 0.454$ for the dlPFC) and between-category comparisons ($t(37) = 0.675$, $p = 0.504$ for the aPFC; $t(37) = 0.416$, $p = 0.680$ for the dlPFC) (Fig. 4B). Therefore, confidence-level-dependent shared representation was not observed during CR trials in the prefrontal cortex despite the fact that the non-learned objects were presented in CR trials as in FR trials. These results further support the idea that the shared prefrontal representation is critically associated with the confidence level of false memories.

3.4. False memory confidence-dependent shared representations were not observed in the posterior regions, including the visual cortex

To test the possibility that false memory confidence also depends on representations in the posterior regions, including the sensory cortex, we subsequently used a whole brain searchlight procedure (Kriegeskorte et al., 2006). By deriving corresponding indices across the whole brain, we found significant differences in the corresponding indices between high-confidence and low-confidence FA trials in the prefrontal regions and anterior temporal cortical regions, but few in the posterior regions (Fig. 5A). These results suggest that false mem-

ory confidence mainly depends on the shared representations in anterior regions, including the prefrontal cortex.

For a closer examination of false-memory confidence-dependent shared representations in the visual cortex, we focused on the individually defined retinotopic early visual cortex (EVC) and object-selective lateral occipital complex (LOC). Consistent with the searchlight results, there were no significant differences in the corresponding indices between FA trials with high confidence and those with low confidence in these visual regions (within-category: $t(30) = 0.168$, $p = 0.868$ for the LOC; $t(30) = 1.164$, $p = 0.254$ for the EVC; between-category: $t(30) = 0.420$, $p = 0.677$ for the LOC; $t(30) = 0.781$, $p = 0.441$ for the EVC) (Fig. 5C). Taken together, these results suggest that the false memory confidence level mainly depends on the reinstatement of high-level components of stored memories rather than the reinstatement of low-level sensory components.

3.5. False memory confidence also depends on the shared representations of the anterior hippocampus

The hippocampus has been considered as a key region for declarative memory storage (Eichenbaum, 2017, 2004, 2000; Manns and Eichenbaum, 2006; Tulving and Markowitsch, 1998) and semantic generalization processes (Shohamy and Wagner, 2008; Zeithamova et al., 2012). Moreover, the anterior hippocampus exhibits stronger functional connectivity with an inferior frontal lobe while the posterior hippocampus shows stronger functional connectivity with the occipital and parietal regions (Libby et al., 2012; Tang et al., 2020). Thus, we examined whether false memory confidence levels also depend on hippocampal representations. We derived the corresponding indices based on the responses in the hippocampus and compared the corresponding indices between FA trials with high confidence and those with low confidence. The corresponding indices derived from the responses of the whole hippocampus during the FA trials with high-confidence were not significantly different from those during the FA trials with low-confidence ($t(34) = 0.808$, $p = 0.425$ for within-category; $t(34) = 0.246$, $p = 0.807$ for between-category) (Fig. 6A).

Given that the anterior and posterior parts of the hippocampus showed differential functional connectivity (Kahn et al., 2008; Libby et al., 2012; Tang et al., 2020) and involvement in memory processes (Collin et al., 2015; Frank et al., 2019; Nadel et al., 2013; Poppenk et al., 2013; Ren et al., 2018; Schlichting et al., 2015), we subsequently undertook a more detailed analysis of the different parts along the anterior-posterior axis of the hippocampus. We divided the hippocampus into four parts with equal lengths along the anterior-posterior axis (Fig. 6B) and derived corresponding indices based on the responses of each part. Remarkably, we found the same tendency of the corresponding indices with those of the prefrontal cortex in the most anterior part of the hippocampus (Fig. 6C). In the most anterior part (0 ~ 25%), corresponding indices during FA trials with high-confidence were significantly greater than those with low-confidence in the within-category comparison ($t(34) = 2.137$, $p = 0.040$) (Fig. 6C). Moreover, the corresponding indices during FA trials with high-confidence in the within-category comparison were significantly greater than those in the between-category comparison ($t(34) = 2.121$, $p = 0.041$) (Fig. 6C). We did not observe any significant differences in the corresponding indices between high- and low-confidence levels ($t(34) = -0.377$, $p = 0.709$ for the HPC (25 ~ 50%); $t(34) = 0.838$, $p = 0.408$ for the HPC (50 ~ 75%); $t(34) = -0.171$, $p = 0.866$ for the HPC (75 ~ 100%)) or between the within-category and between-category comparisons in the posterior parts of the hippocampus ($t(34) = -0.315$, $p = 0.754$ for the HPC (25 ~ 50%); $t(34) = 1.201$, $p = 0.238$ for the HPC (50 ~ 75%); $t(34) = 0.308$, $p = 0.760$ for the HPC (75 ~ 100%)) (Fig. 6C). We also found an identical tendency of the corresponding index when an absolute confidence threshold was used (Supplementary Fig. 3B). However, a one-way ANOVA with hippocampal subregions (0 ~ 25%, 25 ~ 50%, 50 ~ 75%, 75 ~ 100%) as a factor yielded no significant effect in

the corresponding indices for high confidence within-comparison ($F(3, 34) = 1.153$, $p = 0.327$), indicating no clear difference in the shared representations between hippocampal subregions, although there are the prominent confidence-dependent shared representations in the anterior hippocampus (0 ~ 25%). Additionally, we conducted a permutation test to confirm that the different numbers of trials for the high- and low-confidence ratings did not influence the corresponding indices of the most anterior part of the hippocampus. No significant effect of the trial number difference was found in this region (Supplementary Fig. 5A, B). Together, these results suggest that false memory confidence also depends on shared representations with true memory in the anterior hippocampus.

4. Discussion

It is ironic to have a high level of confidence in a false memory that has never been stored. Here, we attempted to discern the neural basis of such confidence in false memories. We found that false memory confidence critically depends on the reinstatement of high-level components of stored memory in the prefrontal regions (Fig. 3). Moreover, when non-learned objects were correctly discriminated as non-learned (correct rejection), confidence-dependent shared representations were not observed (Fig. 4). In previous studies, the overlapping activities of the prefrontal cortex can be interpreted as a reflection of the shared semantic gist but can also be interpreted as a general control process independent of semantic representations (Kouneiher et al., 2009; McGuire and Botvinick, 2010; Wagner et al., 2001). Our findings not only provide direct evidence that the overlapping prefrontal activities reported in prior studies reflect the shared reinstatement of high-level semantic components of stored memory, but also suggest that the degree of the shared representation is critically linked to the confidence level of the false memory. Further, we showed that confidence-dependent shared representations were not observed in the posterior regions, including the sensory cortex. These results demonstrate that false memory confidence is primarily based on high-level semantic components rather than on sensory components shared with a true memory.

Theoretical explanations of false memories suggest that the similarity (Coane et al., 2021; Garoff-Eaton et al., 2007; Gutchess and Schacter, 2012; Slotnick and Schacter, 2004) or source-misattribution (Dodson et al., 2000; Lyle and Johnson, 2006; Stark et al., 2010) is a main factor behind the creation of false memories. In particular, we focus here on the similarity-based false memory represented by the DRM paradigm (Roediger and McDermott, 1995), which is one of the most commonly used paradigms in behavioral experiments to generate false memories (Jou and Flores, 2013; Kim and Cabeza, 2007a; Moritz et al., 2006a; Ruci et al., 2009; Wang et al., 2019). To explain this type of false memory, fuzzy-trace theory proposes that when we store an experience, two parallel traces, a verbatim trace for specific information and a gist trace for general meaning, are stored and that false memories depend on the gist trace based on the similarity between stimuli (Brainerd and Reyna, 2002). Our results provide a direct neural basis for this theory. The shared representations in our results can be interpreted as the gist trace between true and false memories in the fuzzy-trace theory. False memory confidence may depend on the degree to which the prefrontal representation weights gist traces. Moreover, our results further suggest that the high-level semantic gist rather than the perceptual gist is critically linked to the confidence level of the false memory. In correct rejection trials, the gist traces and verbatim traces may be balanced in the prefrontal cortex as well as the visual cortex, allowing discrimination of similar experiences. Global matching models, another theoretical explanation of false memories, propose that the memory is encoded as feature vectors and false memories rely on the degree of the feature overlap between a retrieval probe and stored memory traces. Our results are also consistent with the views of these models in that false memory confidence depends on the overlap of features processed in the prefrontal cortex. Furthermore, we show that not all features have equal weights

and that high-level semantic features are relatively more critical for false memory confidence.

In our results, the difference in the degree of shared representation (corresponding index) between high- and low-confidence levels is prominent in the objects belonging to the same basic-level category (within-category objects) compared to between-category objects (Figs. 3 and 6). This tendency was commonly observed in the aPFC, dlPFC and anterior hippocampus. Given that within-category objects share the same category information and are more closely related, the shared prefrontal representations may reflect a common semantic gist. In addition, although the objects belonging to the same basic-level category also share the perceptual gist, the confidence-dependent shared representation was not observed in the visual cortex for within-category comparison as well as between-category comparison (Fig. 5). Therefore, these results support that false memory confidence is critically based on the degree of the shared semantic gist rather than the perceptual gist.

In the aPFC, a significant difference in the corresponding index between high- and low-confidence levels was also found for the between-category comparisons, although the absolute value of this difference was much smaller than that for the within-category comparisons (Fig. 3). This result is likely a reflection of the degree of the sharing of a higher level of semantic information, such as a superordinate category. In addition, given that the aPFC is considered as a core region related to metacognitive ability (Baird et al., 2013; Molenberghs et al., 2016; Shekhar and Rahnev, 2018), another possibility is that the shared representation in the aPFC reflects the confidence level itself. These possibilities are areas that can be addressed in future work. Nevertheless, our results suggest that the prefrontal representations in the experimental condition of the present study mainly reflect object information (or basic-level category) rather than a superordinate category or confidence level, given that the tendency of confidence-dependent similarity shown in the within-category comparisons was not found in the between-category comparisons in the dlPFC (Fig. 3C) and considering that the corresponding indices with high confidence for the within-category comparisons were significantly greater than the corresponding indices with high confidence for the between-category comparisons in both the aPFC and dlPFC (Fig. 3C).

Prior studies of memory confidence have reported significant difference in neural activities between high- and low-confidence conditions (Bang and Fleming, 2018; De Martino et al., 2013; Henson et al., 2000; Shapiro and Grafton, 2020). However, we did not observe significant difference in the average magnitudes of neural responses between high-confidence and low-confidence FA trials (Supplementary Fig. 4). These discrepancies may arise from differences in measurement way for individual confidence levels. Prior studies usually used two or three levels of confidence rating, while current study used 11-levels of confidence rating. This fine-scale assessment has the advantage of being able to express subtle differences in confidence, but there is also the potential for large inter-individual variability. This large variation in measuring subjective confidence level might cause less clear common distinction between high- and low-confidence levels.

We also found the confidence-dependent shared representations in the anterior part of the hippocampus (Fig. 6). Prior studies have reported confidence-dependent neural responses in the hippocampus (Kim and Cabeza, 2007a; Kuchinke et al., 2013; Ren et al., 2018). Moreover, the anterior hippocampus is thought to be involved in generalization of memory (Frank et al., 2019; Ren et al., 2018; Robin and Moscovitch, 2017). Furthermore, hippocampal subregions exhibited differential functional connectivity during memory encoding; anterior hippocampus shows stronger functional connectivity with inferior frontal and lateral temporal regions while posterior hippocampus shows stronger functional connectivity with the medial and superior frontal lobe and posterior regions including occipital lobe (Cowan et al., 2020; Grady, 2020; Kahn et al., 2008; Libby et al., 2012; Moser and Moser, 1998; Sperling et al., 2003; Strange et al., 2014; Tang et al., 2020; Van Hoesen, 1982). This previous research proposes the possibility

that the anterior hippocampus plays an important role distinct from the role of the posterior hippocampus in memory confidence processing, and this prediction is consistent with our results of false memory confidence (Fig. 6C). It will be interesting to further reveal detailed mechanism how the anterior hippocampus collaborates with the prefrontal regions for the processing of false memory confidence.

5. Conclusions

We found that the confidence level for false memories, which have never been stored in the brain is critically based on the prefrontal re-statement of high-level components of the true memory. Our findings provide an integrative view of the theoretical explanations of false memories with the neural bases underlying the explanations and further provide neural targets for the treatment of delusional disorders.

Data and code availability

The behavioral and neuroimaging data are not able to be made openly available because of the need for approval from the Institutional Review Boards. Sharing and reuse of data require a reasonable request by sending an e-mail to the corresponding author with the name of P.I., affiliation, and purpose of the study, and approval from the Institutional Review Boards. The code used in this study is available at this link (<https://memory.kaist.ac.kr/codes/>).

Declaration of Competing Interests

The authors declare no competing interests.

Credit authorship contribution statement

Taehyun Kim: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Inho Shin:** Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Sue-Hyun Lee:** Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing.

Data Availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2022.119597.

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